Area 4: Power Transmission

PE Mechanical: Machine Design & Materials (2025 specification)

Learn each subsection, then take its quiz in the simulator.

Original study material. Not NCEES, Testmasters or PPI content. ★ = focus area.

Contents

4.A Gears and Gear Trains
4.B Bearings
4.C Belts, Chains, Clutches, Brakes, Power Screws
4.D Shafts and Keys
4.E Motors and Engines

4.A Gears and Gear Trains

1. Key concepts

- Geometry: diametral pitch P = N/d (teeth/in) or module m = d/N (mm). Center distance = $(d_1 + d_2)/2$. Mating gears must share P (or m) and the pressure angle.
- Forces (spur): $W_t = 33{,}000hp/V$ (V in ft/min), radial $W_r = W_t \cdot \tan\varphi$. Helical: axial $W_a = W_t \cdot \tan\psi$.
- Train value = product of driver teeth / product of driven teeth. Planetary: use the Willis formula $(n_L - n_A)/(n_F - n_A) = e$.
- Bending (Lewis): $\sigma = K_v W_t P/(FY)$. Wear (contact) uses the AGMA equations.

2. Essential formulas

Name	Formula	Variables / units
Pitch	$P = N/d\ ;\ m = d/N$	
Pitch-line velocity	$V = \pi dn/12\ (\mathrm{ft/min})$	
Tangential load	$W_t = 33{,}000\ hp/V$	
Radial	$W_r = W_t \tan\varphi$	
Helical axial	$W_a = W_t \tan\psi$	
Lewis	$\sigma = K_v W_t P/(FY)$	
Planetary	$(n_{ring} - n_{arm})/(n_{sun} - n_{arm}) = -N_{sun}/N_{ring}$	

3. Handbook search keywords

diametral pitch · gear forces · Lewis bending · gear train value · planetary gear train

4. Worked examples

Example 1: Spur gear forces
An 18T pinion with P = 6 and φ = 20° transmits 5 hp at 1,150 rpm. Find W_t and W_r.

1. d = 18/6 = 3 in; V = π(3)(1,150)/12 = 903 ft/min.
2. $W_t = 33{,}000(5)/903 = 182.7$ lb; $W_r = 66.5$ lb.

Answer: $W_t \approx 183$ lb, $W_r \approx 66$ lb

5. Common traps

- Using V in ft/s.
- Mixing up driver and driven teeth in the train value.
- Using the pressure angle for the helical thrust (use the helix angle).

4.A Gears and Gear Trains: one-page cheat summary

- ► $P = N/d$; $C = (N_1 + N_2)/(2P)$
- ► $V = \pi dn/12$
- ► $W_t = 33{,}000hp/V$
- ► $W_r = W_t \tan\varphi$; $W_a = W_t \tan\psi$
- ► Planetary: Willis formula

Formulas

Pitch	$P = N/d$; $m = d/N$
Pitch-line velocity	$V = \pi dn/12$ (ft/min)
Tangential load	$W_t = 33{,}000\ hp/V$
Radial	$W_r = W_t \tan\varphi$
Helical axial	$W_a = W_t \tan\psi$
Lewis	$\sigma = K_v W_t P/(FY)$
Planetary	$(n_{ring} - n_{arm})/(n_{sun} - n_{arm}) = -N_{sun}/N_{ring}$

Search keywords

diametral pitch, gear forces, Lewis bending, gear train value, planetary gear train

Top traps

- Using V in ft/s.
- Mixing up driver and driven teeth in the train value.
- Using the pressure angle for the helical thrust (use the helix angle).

4.B Bearings

1. Key concepts

- Rolling bearings: $L_{10} = (C/P)^a \times 10^6$ revolutions, with a = 3 for ball and 10/3 for roller bearings. Hours = rev/(60n).
- Equivalent load P = XF_r + YF_a when F_a/F_r > e. Required rating C = P(L_D/L_R)^(1/a).
- Reliability adjustment (Weibull): for R ≠ 0.90, the life scales with [ln(1/R)/ln(1/0.9)]^(1/b).
- Journal bearings: Sommerfeld number S = (r/c)²μN/P; the Petroff torque estimates friction.

2. Essential formulas

Name	Formula	Variables / units
Rating life	$L_{10} = (C/P)^a \times 10^6$ rev	
Hours	$L_h = L_{10}/(60n)$	
Equivalent load	$P = XF_r + YF_a$	
Required C	$C = P(L_h 60n/10^6)^{(1/a)}$	
Sommerfeld	$S = (r/c)^2 \mu N/P$	

3. Handbook search keywords

bearing L10 life · dynamic load rating · equivalent radial load · Sommerfeld number

4. Worked examples

Example 1: Roller bearing life
Roller bearing with C = 15,000 N, P = 4,000 N, n = 3,600 rpm. Find L_{10} in hours.

1. L_{10} = (15,000/4,000)^(10/3) × 10⁶ rev = 81,928,982 rev.
2. Hours = rev/(60 × 3,600) = 379 h.

Answer: 379 h

5. Common traps

- Using a = 3 for roller bearings.
- Forgetting the 60 min/h.
- Using the radial load alone when thrust is present.

4.B Bearings: one-page cheat summary

- ► $L_{10} = (C/P)^a\ 10^6$ rev
- ► a = 3 ball, 10/3 roller
- ► $P = XF_r + YF_a$
- ► $h = rev/60n$

Formulas

Rating life	$L_{10} = (C/P)^a \times 10^6$ rev
Hours	$L_h = L_{10}/(60n)$
Equivalent load	$P = XF_r + YF_a$
Required C	$C = P(L_h 60n/10^6)^{(1/a)}$
Sommerfeld	$S = (r/c)^2 \mu N/P$

Search keywords

bearing L10 life, dynamic load rating, equivalent radial load, Sommerfeld number

Top traps

- Using a = 3 for roller bearings.
- Forgetting the 60 min/h.
- Using the radial load alone when thrust is present.

4.C Belts, Chains, Clutches, Brakes, Power Screws

1. Key concepts

- Flat belt: $T_1/T_2 = e^{(\mu\theta)}$ at impending slip (V-belts use $\mu/\sin(\beta/2)$). Power = $(T_1 - T_2)V$.
- Chain: V = NpN_rpm/12 (ft/min). Chordal action is reduced by using more sprocket teeth.
- Disk clutch (uniform wear): $T = N\mu F(D + d)/4$. Uniform pressure gives slightly more torque. Band brake: $T = (T_1 - T_2)r$ with $T_1/T_2 = e^{(\mu\theta)}$.
- Square-thread power screw: T_raise = (Wd_m/2)(l + πμd_m)/(πd_m − μl) + collar torque Wμ_cd_c/2. It is self-locking if πμd_m > l. Efficiency = Wl/(2πT).

2. Essential formulas

Name	Formula	Variables / units
Belt	$T_1/T_2 = e^{(\mu\theta)}$; $hp = (T_1 - T_2)V/33{,}000$	
Open-belt length	$L = 2C + \pi(D + d)/2 + (D - d)^2/(4C)$	
Clutch (uniform wear)	$T = N\mu F(D + d)/4$	
Screw raise	$T = (Wd_m/2)(l + \pi\mu d_m)/(\pi d_m - \mu l)$	
Efficiency	$e = Wl/(2\pi T)$	

3. Handbook search keywords

belt tension ratio · band brake · disk clutch uniform wear · power screw torque · chain velocity

4. Worked examples

Example 1: Flat belt tensions
A belt transmits 5 hp from a 6 in pulley at 1,150 rpm, θ = 180°, μ = 0.3. Find T_1.

1. Net force $T_1 - T_2 = T/r = (63{,}025 \cdot 5/1{,}150)/3 = 91.3$ lb.
2. $e^{(\mu\theta)} = e^{(0.3\pi)} = 2.566$; $T_1 = F \cdot e/(e - 1) = 149.7$ lb.

Answer: 150 lb

5. Common traps

- Using θ in degrees in $e^{(\mu\theta)}$.
- Forgetting the collar friction on a power screw.
- Using uniform pressure when the question says uniform wear.

4.C Belts, Chains, Clutches, Brakes, Power Screws: one-page cheat summary

- ► $T_1/T_2 = e^{(\mu\theta)}$, θ in rad
- ► $hp = (T_1 - T_2)V/33{,}000$
- ► Clutch: $N\mu F(D + d)/4$
- ► Screw: self-locking if $\pi\mu d_m > l$

Formulas

Belt	$T_1/T_2 = e^{(\mu\theta)}$; $hp = (T_1 - T_2)V/33{,}000$
Open-belt length	$L = 2C + \pi(D + d)/2 + (D - d)^2/(4C)$
Clutch (uniform wear)	$T = N\mu F(D + d)/4$
Screw raise	$T = (Wd_m/2)(l + \pi\mu d_m)/(\pi d_m - \mu l)$
Efficiency	$e = Wl/(2\pi T)$

Search keywords

belt tension ratio, band brake, disk clutch uniform wear, power screw torque, chain velocity

Top traps

- Using θ in degrees in e^(μθ).
- Forgetting the collar friction on a power screw.
- Using uniform pressure when the question says uniform wear.

4.D Shafts and Keys

1. Key concepts

- Design steps: loads → shear and moment diagrams → critical sections (shoulders, keyways) → fatigue size → check deflection, slopes and critical speed.
- DE-Goodman with rotating bending (M_a) and steady torque (T_m): $d = \{16n/\pi\,[2K_{fM_a}/S_e + \sqrt{3}K_{fsT_m}/S_{ut}]\}^{(1/3)}$.
- Critical speed: $\omega = \sqrt{(g/\delta_{st})}$ for a single disk (Rayleigh for several). Keep the operating speed well away from it.
- Flywheel: $\Delta E = C_s I\omega^2$, with $C_s = (\omega_{max} - \omega_{min})/\omega_{avg}$.

2. Essential formulas

Name	Formula	Variables / units
DE-Goodman diameter	$d = \{16n/\pi\,[2K_{fM_a}/S_e + \sqrt{3}K_{fsT_m}/S_{ut}]\}^{(1/3)}$	
Static torsion	$d = (16Tn/(\pi S_{sy}))^{(1/3)}$	
Critical speed	$n_c = (30/\pi)\sqrt{(g/\delta)}$	$g = 386$ in/s²
Flywheel	$\Delta E = C_s I\omega^2$	

3. Handbook search keywords

shaft design DE-Goodman · critical speed · flywheel coefficient of fluctuation · key

4. Worked examples

Example 1: Critical speed
A 250 lb disk sits at the center of a Ø1.5 in steel shaft ($E = 30\times10^6$ psi) on bearings 48 in apart. Find n_c.

1. $I = \pi(1.5)^4/64 = 0.2485$ in⁴; $\delta = WL^3/48EI = 0.0773$ in.
2. $n_c = (30/\pi)\sqrt{(386.1/\delta)} = 675$ rpm.

Answer: 675 rpm

5. Common traps

- Using $g = 32.2$ ft/s² with δ in inches.
- Leaving out K_f at shoulders.
- Applying the mean-torsion term to the alternating stress.

4.D Shafts and Keys: one-page cheat summary

- ► DE-Goodman d formula
- ► $n_c = (30/\pi)\sqrt{(g/\delta)}$
- ► $\Delta E = C_s I \omega^2$
- ► Check slopes at bearings

Formulas

DE-Goodman diameter	$d = \{16n/\pi\ [2K_f M_a/S_e + \sqrt{3}K_{fs}T_m/S_{ut}]\}^{(1/3)}$
Static torsion	$d = (16Tn/(\pi S_{sy}))^{(1/3)}$
Critical speed	$n_c = (30/\pi)\sqrt{(g/\delta)}$
Flywheel	$\Delta E = C_s I \omega^2$

Search keywords

shaft design DE-Goodman, critical speed, flywheel coefficient of fluctuation, key

Top traps

- Using g = 32.2 ft/s² with δ in inches.
- Leaving out K_f at shoulders.
- Applying the mean-torsion term to the alternating stress.

4.E Motors and Engines

1. Key concepts

- Induction motor: synchronous speed n_s = 120f/p (p = poles). Slip s = (n_s − n)/n_s.
- Three-phase input: P = √3·V·I·pf. Output = η × input. T = 63,025hp/n.
- Engines: brake power = 2πNT. Thermal efficiency = output/fuel energy. BSFC = fuel rate/brake power.

2. Essential formulas

Name	Formula	Variables / units
Sync speed	$n_s = 120f/p$	
Slip	$n = n_s(1 - s)$	
3-phase power	$P = \sqrt{3} \cdot V \cdot I \cdot pf$	
Line current	$I = hp \cdot 746/(\sqrt{3} \cdot V \cdot \eta \cdot pf)$	

3. Handbook search keywords

induction motor slip · synchronous speed · three phase power · motor efficiency

4. Worked examples

Example 1: Motor current
A 25 hp, 460 V three-phase motor has η = 0.90 and pf = 0.85. Find the line current.

1. I = 25(746)/(√3·460·0.90·0.85) = 30.6 A.

Answer: 30.6 A

5. Common traps

- Using poles/2 incorrectly (n_s = 120f/p with p = total poles).
- Forgetting efficiency when going from output to input power.

4.E Motors and Engines: one-page cheat summary

- ► $n_s = 120f/p$
- ► $n = n_s(1 - s)$
- ► $P = \sqrt{3}VI \cdot pf$
- ► $T = 63{,}025hp/n$

Formulas

Sync speed	$n_s = 120f/p$
Slip	$n = n_s(1 - s)$
3-phase power	$P = \sqrt{3} \cdot V \cdot I \cdot pf$
Line current	$I = hp \cdot 746/(\sqrt{3} \cdot V \cdot \eta \cdot pf)$

Search keywords

induction motor slip, synchronous speed, three phase power, motor efficiency

Top traps

- Using poles/2 incorrectly ($n_s = 120f/p$ with p = total poles).
- Forgetting efficiency when going from output to input power.